Real-Time Road Object Detection on Duckiebot Using YOLO-Based Vision Perception

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INTRODUCTION

Autonomous mobile robots operating in urban environments require reliable and real-time perception of surrounding objects to ensure safe navigation and decision-making. In the Duckietown ecosystem, Duckiebot serves as a representative autonomous vehicle platform for research and education, where vision-based perception plays a critical role in understanding road scenes.

This work focuses on deploying a YOLO-based object detection system on Duckiebot to recognize common road-level objects, such as obstacles, vehicles, and pedestrians (duckies). YOLO (You Only Look Once) is a state-of-the-art single-stage object detection framework known for its high inference speed and strong detection performance, making it well-suited for resource-constrained embedded robotic platforms.

We integrate the YOLO model into the Duckiebot perception pipeline and evaluate its real-time performance under on-board computation constraints. The proposed system demonstrates effective object detection capabilities in Duckietown road scenarios, providing reliable semantic information for downstream tasks such as obstacle avoidance and autonomous navigation.

Dataset

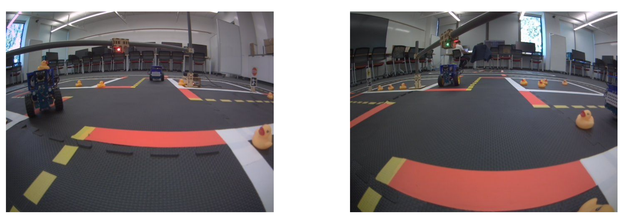

Figure 1: Examples of image from dataset

Class	Number of Samples
duckie	1116
duckiebot	340
red_traffic_light	241
stop_sign	154
3_way_sign	104
green_traffic_light	50

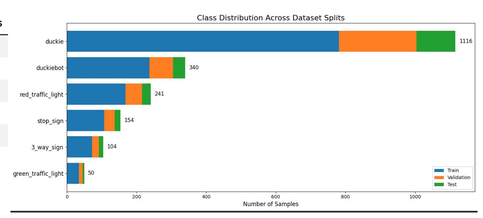

Figure 2: Class Distribution of the Dataset.

We construct a custom object detection dataset using images collected from the on-board Duckiebot camera in Duckietown environments. The dataset includes six object classes: duckie, duckiebot, red traffic light, green traffic light, stop sign, and 3-way intersection sign.

All images are manually annotated using Roboflow and converted into YOLO format. Data augmentation increases the dataset to 1,438 images. The dataset is split into training, validation, and test sets with a ratio of 70:20:10.

METHODOLOGY

This study investigates the deployment of YOLO-based object detection for real-time road perception on the Duckiebot platform. A YOLO model is trained using a customized dataset. The trained model is integrated into the Duckiebot perception pipeline and evaluated under real-time operating conditions. In contrast to previous Duckietown approaches, this work adopts the more recent YOLOv11 model, which has not been used in this context before.

YOLO (You Only Look Once) formulates object detection as a single-stage regression problem, directly predicting bounding boxes and class probabilities from input images in a single forward pass. Given an input image, the model outputs a set of bounding boxes with corresponding confidence scores, enabling efficient and low-latency inference suitable for real-time on-board deployment on the Duckiebot.

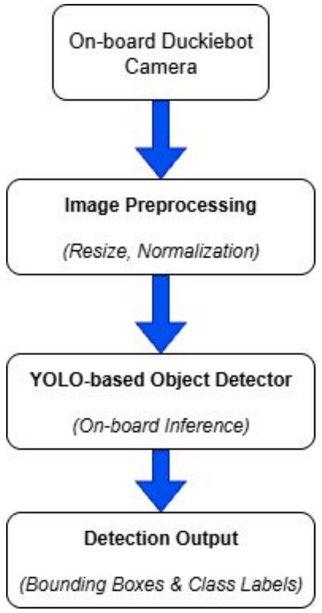

Figure 3: YOLO-based real-time object detection pipeline on Duckiebot.

RESULTS

Our model demonstrates reliable real-time object detection on images captured by the Duckiebot camera. Quantitative results are shown in Figure 4, and qualitative detection examples can be viewed by scanning the QR code in Figure 5.

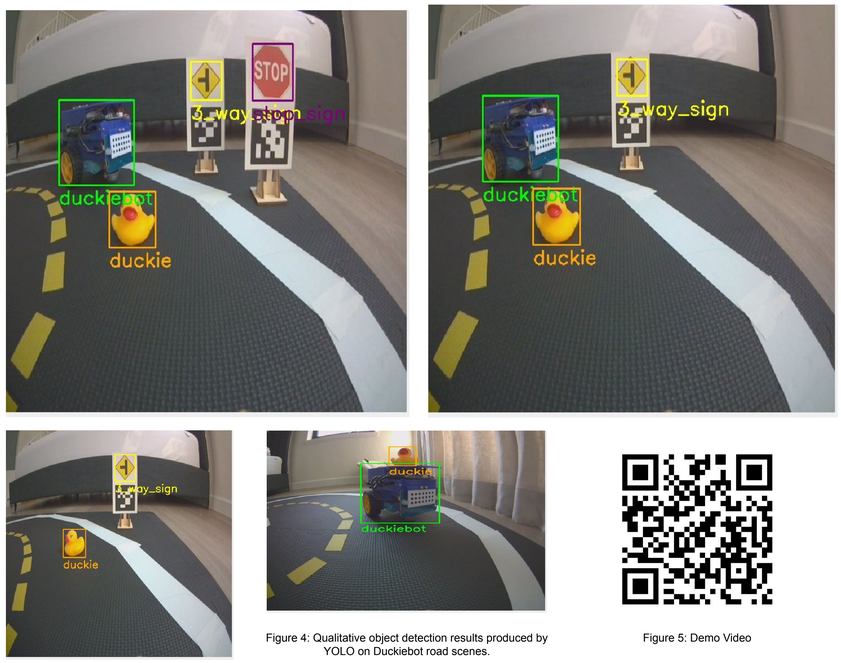

Figure 4: Qualitative object detection results produced by YOLO on Duckiebot road scenes.

Figure 5: Demo Video

Overall Metrics

Metric	Value
Precision	0.8931
Recall	0.7991
mAP@50	0.9210
mAP@50:95	0.6420

Per-class Metrics

Class	AP@50	AP@50:95
3_way_sign	0.9125	0.6201
duckie	0.9000	0.6833
duckiebot	0.9859	0.7751
green_traffic_light	0.9950	0.6259
red_traffic_light	0.7377	0.4567
stop_sign	0.9950	0.6911

Figure 6: Detection Performance on the Test Set。

CONCLUSION & FUTURE WORK

In this study, we present a YOLOv11-based object detection system deployed on the Duckiebot platform for real-time road perception in Duckietown environments. Using a custom dataset collected from the on-board camera, the proposed approach demonstrates effective and reliable object detection under embedded computational constraints. Although real-time performance is limited by the available on-board resources, the trained YOLO model produces accurate and consistent detection results, highlighting the effectiveness of task-specific dataset construction for robotic perception.

Future work will focus on optimizing inference efficiency under limited hardware resources through model compression and acceleration techniques. We also plan to incorporate quantitative evaluation metrics, such as inference latency and detection accuracy, to provide a more comprehensive performance analysis. In addition, the detection results will be integrated with downstream modules, including obstacle avoidance and autonomous navigation, to enable closed-loop decision-making on Duckiebot. Finally, we aim to expand the dataset to more complex road scenarios and object categories to improve robustness and generalization in real-world deployments.

REFERENCE

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